(a) 1 only (b) 2 and 4 only
(c) 1 and 3 (d) 2, 3 and 4

Ans. (c): The Reserve Bank carries out the task of controlling the quantity and direction of credit created by commercial banks. He uses quantitative and qualitative measures to achieve this objective. When the RBI is looking to increase the quantity of money in circulation, it purchases government securities from commercial banks and institutions.

63. Consider the following statements :
The price of any currency in international market is decided by the-
1. World Bank.
2. Demand for goods/services provided by the country concerned.
3. Stability of the government of the concerned country.
4. Economic potential of the country in question.

Which of the statements given above are correct?
(a) 1, 2, 3 and 4 (b) 2 and 3 only
(c) 3 and 4 only (d) 1 and 4 only

Ans. (b): Money supply increases when money flows out of the RBI. The purchase of Government securities from the public by the central bank leads to the transfer of money to the public, thereby leading to an increase in money supply in the economy. Similarly, borrowing by the government from the central bank leads to an increased money supply.
However, the deposit of currency in commercial banks by the public doesn't increase money supply as the money just gets transferred from the public to commercial banks.

64. The basic aim of Lead Bank Scheme is that-
(a) Big banks should try to open offices in each district.
(b) There should be stiff competition among the various nationalized banks.
(c) Individual banks should adopt particular districts for intensive development.
(d) All the banks should make intensive efforts to mobilize deposits.

Ans. (c): The basic aim of the Lead Bank Scheme is that stiff competition among the various nationalized banks and big banks should try to open offices in each district.

65. Which of the following is/are among the noticeable features of the recommendations of the Thirteenth Finance Commission?
1. A design for the Goods and Service Tax, and a compensation package linked to adherence to the proposed design.
2. A design for a creation of lakhs of jobs in the next ten years in consonance with India's demographic dividend.
3. Devolution of a specified share of central taxes to local bodies as grants.

Select the correct answer using the codes given below:
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (c): The 13th Finance Commission report has an elaborate design for the GST. Statement (1) is correct while Statement (2) is incorrect. The report basically discusses the improved implementation of the existing schemes. The statement (3) is correct. The report in its para 10.144 mentions: "Taking into account the demand of local bodies that they be allowed to benefit from the buoyancy of central taxes and the constitutional design of supplementing the resources of Panchyayats and Municipalities through grant-in-aid, we recommend that local bodies be transferred a percentage of the divisible pool of taxes (over and above the share of the states) as stipulated by as, after converting this share to grant-in-aid under article 275".

• Foreign Trade of India, Economic Organizations & Share Market

66. Which of the following would include Foreign Direct Investment in India ?
1. Subsidiaries of companies in India.
2. Majority foreign equity holding in Indian companies.
3. Companies exclusively financed by foreign companies.
4. Portfolio investment.

Select the correct answer using the codes given below :
(a) 1, 2, 3 and 4 (b) 2 and 4 only
(c) 1 and 3 only (d) 1, 2 and 3 only

Ans. (d): Foreign Direct Investment (FDI) is a direct investment into production or business in a country by an individual or company of another country, either by buying a company in the target country or expanding operations of an existing business in that country. Foreign direct investment is in contrast to portfolio investment, which is a passive investment in the securities of another country such as stocks and bonds.

SCIENCE

• Physics

67. A person stood alone in a desert on a dark night and wanted to reach his village which was situated 5 km east of the point where he was standing. He had no instruments to find the direction but he located the polestar. The most convenient way now to reach his village is to walk in the –
(a) direction facing the polestar
(b) direction opposite to the polestar
(c) direction keeping the polestar to his left
(d) direction keeping the polestar to his right

Ans. (c): Pole star's position with respect to rotating earth never changes and it always points to the north direction. Thus, if the person keeps the pole star to his left, he can be sure of walking in the eastern direction in which his village is located. Polestar was used widely for navigation in ancient and medieval times due to its stable position.

68. Consider the following statements :
If there were no phenomenon of capillarity-
1. It would be difficult to use a kerosene lamp.
2. One would not be able to use a straw to consume a soft drink.
3. The blotting paper would fail to function.
4. The big trees that we see around would not have grown on the Earth.

Which of the statements given above are correct?
(a) 1, 2 and 3 only (b) 1, 3 and 4 only
(c) 2 and 4 only (d) 1, 2, 3 and 4

Ans. (b): If capillary phenomena were not present, the use of the Kerosene lamp would have been difficult, the blotting paper would have failed to function and the large trees that we see all around would not grow on the earth because capillarity is fundamentally responsible for all these phenomena.

69. Graphene is frequently in news recently. What is its importance?
1. It is a two-dimensional material and has good electrical conductivity.
2. It is one of the thinnest but strongest materials tested so far.
3. It is entirely made of silicon and has high optical transparency.
4. It can be used as 'conducting electrodes' required for touch screens, LCDs and organic LEDs.

Which of the statements give above are correct?
(a) 1 and 2 only (b) 3 and 4 only
(c) 1, 2 and 4 only (d) 1, 2, 3 and 4

Ans. (c): "Graphene" is a two-dimensional material and has good electrical conduction. It is one of the thinnest but strongest materials tested so far. It can be used for conducting electrodes required for touch screens, LCDs and organic LEDs.

70. Which of the following is /are cited by the scientists as evidence/evidences for the continued expansion of universe ?
1. Detection of microwaves in space.
2. Observation of red shift phenomenon in space.
3. Movement of asteroids in space.
4. Occurrence of supernova explosions in space.

Select the correct answer using the codes given below:
(a) 1 and 2
(b) 2 only
(c) 1, 3 and 4
(d) None of the above can be cited as evidence

Ans. (a): Scientists cite the observation of a red shift phenomenon in space and the presence of microscopic waves in space as evidence of the continued expansion of the universe. In addition, supernova explosions in space and the movement of asteroids etc. are not used as an extension of the universe.

71. Electrically charged particles from space travelling at speeds of several hundred km/sec can severely harm living beings if they reach the surface of the Earth .What prevents them from reaching the surface of the Earth ?
(a) The Earth's magnetic field diverts them towards its poles.
(b) Ozone layer around the Earth reflects them back to outer space.
(c) Moisture in the upper layers of atmosphere prevents them from reaching the surface of the Earth.
(d) None of the statements (a), (b) and (c) given above is correct.

Ans. (a) : Electric charge particles travelling at a speed of several hundred km/sec in space cannot reach the surface of the earth because the Earth's magnetic force turns them towards the pole.

• Chemistry

72. Which one of the following sets of elements was primarily responsible for the origin of life on the Earth ?
(a) Hydrogen, Oxygen, Sodium
(b) Carbon, Hydrogen, Nitrogen
(c) Oxygen, Calcium, Phosphorus
(d) Carbon, Hydrogen, Potassium

Ans. (b): Carbon, Hydrogen, Nitrogen are basically responsible for the origin of life on Earth. Because of these primary gases, the necessities of life, availability of water, food and environment etc. are ensured.

73. Recently, there has been a concern over the short supply of group of elements called 'rare earth metals'. Why ?
1. China, which is the largest producer of these elements, has imposed some restrictions on their export.
2. Other than China, Australia, Canada and Chile, these elements are not found in any country.
3. Rare earth metals are essential for the manufacture of various kinds of electronic items and there is a growing demand for these elements.

Which of the statements given above is/are correct?
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (c): Origin of life started with the formation of simple organic molecules like nucleoside. The building blocks of nucleoside are Carbon, Hydrogen, Oxygen, Nitrogen and Phosphorus. Carbon is the most important of these because of its capability to form a wide variety of compounds.

74. What is the role of ultraviolet (UV) radiation in the water purification systems.
1. It inactivates/kills the harmful microorganisms in water.
2. It removes all the undesirable odours from the water.
3. It quickens the sedimentation of solid particles, removes turbidity and improves the clarity of water.

Which of the statements given above is/are correct?
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3